

glucose

A simple carbohydrate, or sugar, made by the process of photosynthesis and then used by cells as a source of energy.

gravity

A force that acts between all masses and which tends to pull them together.

habitat

The place where organisms live; every habitat has specific conditions, such as supply of water, range of temperatures, and amount of light.

half-life

The period of time that it takes for a sample of a radioactive element to halve in mass by decaying into other elements.

hormone

A chemical messenger that travels through the bloodstream to control certain life processes; hormones include epinephrine, insulin, and estrogen.

hydrocarbon

A compound composed largely, if not entirely, from hydrogen and carbon.

immiscible

A property where two liquids will not mix with each other because their molecules push away from each other.

indicator

A substance that changes color with pH, the measure of acidity.

induction

The process by which the energy of a moving conductor is converted into an electrical current when it passes through a magnetic field.

inertia

A mass's resistance to changing its state of motion.

insulation

A material with the function of stopping heat moving from a warm object to a colder one; animal insulation, such as hair or blubber, is used to save energy.

interference

The mixing of two or more light waves to produce new, different ones.

invertebrate

An animal with no backbone. Most animals are invertebrates, but are nevertheless not all closely related.

ion

An atom or a molecule that has lost or gained an electron and thus carries a positive or negative charge.

isotope

One of two or more forms of atom all with the same number of protons—and so belonging to the same element—but with varying numbers of neutrons.

keratin

A protein used by vertebrates to cover their bodies; feathers, hair, nails, claws, horns, and reptile scales are all made of keratin.

longitudinal

A wave that is made up of compressions and expansions of a medium.

main sequence star

An average star, like our Sun.

mass

A property of an object that allows it to have weight and be acted on by forces.

matter

Anything that has mass and occupies space.

membrane

A thin layer that surrounds a cell or other body structure; the layer is semipermeable, so only certain substances can cross it.

metabolism

The name used for all processes that support life that take place in a living body; catabolism is all the processes that break things into simpler substances; anabolic processes build simple substances into complex ones.

metal

An element that is likely to react by losing electrons, forming a cation; metals are generally shiny, heavy solids.

micrometer (μm)

A millionth of a meter.

microtubule

A fine fiber of protein that runs through the cytoplasm of a cell and is used to haul larger items around.

mixture

A collection of two or more substances mixed together but which are not chemically connected.

mole

A unit of quantity used to count huge numbers of objects, such as atoms and molecules; for example, one mole of hydrogen atoms is 6.0221415×10^{23} atoms.

molecule

Two or more atoms that are bonded together; the molecule is the smallest unit of a compound; breaking it up into simpler units would destroy the compound.

momentum

The product of the speed of an object and its mass.

nanometer (nm)

A billionth of a meter.

neutron

A neutral particle located in the nuclei of most atoms.

nucleus (plural: nuclei)

The central core of something. An atomic nucleus contains

protons and neutrons, while a cell's nucleus contains DNA.

nutrient

A substance that is useful for life as a source of energy or as raw material.

octet

A collection of eight things.

orbit

The path of one mass around another mass under the influence of gravity.

organelle

A structure inside a cell that performs a certain task in the cell's metabolism.

organism

A living thing.

oscillation

A regular vibration around a fixed point.

oxidation

The loss of electrons by an atom, ion, or molecule.

phloem

The vascular tissue that carries sugar fuel around a plant.

pigment

A chemical substance that colors an object.

plasma

A high-energy state of matter where the atoms of a gas have been ripped into their constituent parts.

polarity

Relating to an object, such as a magnet, that has two opposite ends or poles.

polymer

A long chainlike molecule made up of smaller molecules connected together.

precipitation

A solid or liquid that falls from a cloud. Rain, snow, sleet, and hail